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title: "Centaurus A AGN F_U_Bi_i Curves — 7-Point Gamma Sweep with Jet Modulation"
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cvw: "v2.0.0"
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crosslinks: [PAPER_989, PAPER_993, PAPER_930]
calibration: {M_BH: "5.5e7 Msun", a_spin: 0.70, B_gauss: 3000, gamma_points: 7}
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PAPER_991: Centaurus A AGN F_U_Bi_i Curves

Abstract

We compute numerical F_{U,Bi_i} curves for Centaurus A ($M_{\mathrm{BH}} = 5.5 \times 10^7\, M_\odot$, $a = 0.70$, $B = 3000$ G) across 7 linewidth values $\Gamma \in \{0.01, 0.05, 0.1, 0.5, 1.0, 5.0, 10.0\}$ THz.

1. Jet Modulation Factor

$$M_{\mathrm{jet}}(\Gamma) = 1 + A_{\mathrm{jet}} \cdot \exp\left(-\frac{(\Gamma - \Gamma_0)^2}{2\sigma_\Gamma^2}\right)$$

with $A_{\mathrm{jet}} = 1.5$, $\Gamma_0 = 2\pi \times 0.1$ THz, $\sigma_\Gamma = 0.08 \times 2\pi$ THz.

2. Jet Power

$$P_{\mathrm{jet}}(\Gamma) = \frac{B^2}{8\pi} \left(\frac{r_H}{c}\right)^2 a^2 c \cdot M_{\mathrm{jet}}(\Gamma)$$

3. Combined F_U_Bi_i at Horizon

$$F_{U,\mathrm{Bi}_i}^{\mathrm{CenA}}(\Gamma) = U_g(r_H) - U_b(r_H) + P_{\mathrm{jet}}(\Gamma) \times 10^{-45}$$

The jet power coupling at 10^{-45} ensures dimensional consistency with the acceleration-scale F_U_Bi_i.

4. Implementation

File: `fubi_inside_outside.py`, class `CentaurusAFUBiCurves`. CP4 class #575.